

Software Development Workshop II

On the Course Project

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Timeline

- Project Release (Mar 25)
- Group Selection (DDL: Apr 1)
- Submission of the Software Development Specification (DDL: Apr 22)
- Submission of the First draft Implementation (DDL: May 13)

Timeline

- Submission of Final (V2) Implementation and Report (DDL: May 20)
 - In the report, please specify the overall percentage of work each person contributed to the project
- Presentation (May 25-29)
 - Presentation is conducted on the lab hours of each section

Note: The templates for the SDS and the Report are provided on iSpace

Grouping

- Each group can have 6 members (or 5 if you belong to the last few groups).
- Find your teammates and register to a group on iSpace.
- The deadline for group choice is April 1. If you don't register before the deadline, we will assign you a group that is not full.
- Please decide a leader for your group. The leader acts as the contact person, and has no extra responsibility.

Project Format

- Objective
 - Data processing on real-world datasets
- 5 datasets are provided to you
 - Explore, analyze, and derive insights from them
- Choose your own focus
 - You will pick one of the 5 datasets
 - You will decide what to do with the dataset

Components of your Work

Your work must include the following components.

1: Data Exploration and Cleaning

- Load the dataset into a Python environment (e.g., Jupyter Notebook).
- Perform initial exploration to understand the structure, variables, and potential issues (e.g., missing values, duplicates, inconsistencies).
- Clean the dataset by handling missing values, removing duplicates, and correcting inconsistencies.

2: Data Analysis

- Perform exploratory data analysis (EDA) to uncover patterns, trends, and relationships in the data.
- Use statistical methods and visualizations to support your findings.
- Generate at least insights from the dataset.

3: Data Visualization

- Create meaningful visualizations to present your findings.
- Ensure visualizations are well-labeled and easy to interpret.

4: Advanced Tasks (Optional)

- Building a machine learning model (e.g., classification, regression, clustering).
- Performing time-series analysis or geospatial analysis.

5 Datasets

You must pick one from the following 5 datasets.

1: COVID-19 Dataset

- Contains global COVID-19 data, including cases, deaths, recoveries, and vaccination rates.
- **Use Cases:** Data cleaning, time-series analysis, trend visualization, and regional comparisons.
- **Source:** [Our World in Data](#)

2: FIFA Player Dataset

- Contains detailed information about FIFA players, including ratings, positions, and skills.
- **Use Cases:** Data cleaning, player performance analysis, team building, and visualization.
- **Source:** [Kaggle FIFA Dataset](#)

3: Pokemon Dataset

- Contains information about Pokemon, including stats, types, and abilities.
- **Use Cases:** Data cleaning, type analysis, and visualization.
- **Source:** [Kaggle Pokemon Dataset](#)

4: IMDB Movies Dataset

- Consists of movies released on or before July 2017. Data points include cast, crew, plot keywords, budget, revenue, posters, release dates, languages, production companies, countries, TMDB vote counts and vote averages.
- **Use Cases:** Data cleaning, genre analysis, rating trends, and visualization.
- **Source:** [Kaggle IMDB Dataset](#)

5: Human Activity Recognition Dataset

- Contains data and labels from IMU sensors in mobile phones, including sensor signals, training set, testing set and corresponding labels.
- **Use Cases:** Data cleaning, model training, and visualization.
- **Source:** [UCI HAR Dataset](#)

Presentation

- Presentation length: 8 - 10 minutes for presentation
- Everyone in the group must speak.
- It's recommended that every one introduce the part that they do.
- Some people can speak longer than others, but everyone must speak.

Presentation

- If your English is not so good, you can make your slides good.
- In your PPT, please put your name on the slides on which you speak.
- Please show the page number on your slides.
- Right before your presentation, please give the teachers 2 printed copies of your PPT.
 - Print at least 4 slides on each page!

Grading

- You will get full mark for the first draft SDS and the first draft implementation if you complete the submission. (5%)
- The rest of the assessments are based on your presentation and your final submission. (95%)
 - Please refer to the rubrics for details.
 - You will get full mark for Teamwork if you make reasonable contribution to the project.
 - For example, about 20% of contribution if your group size is 5.